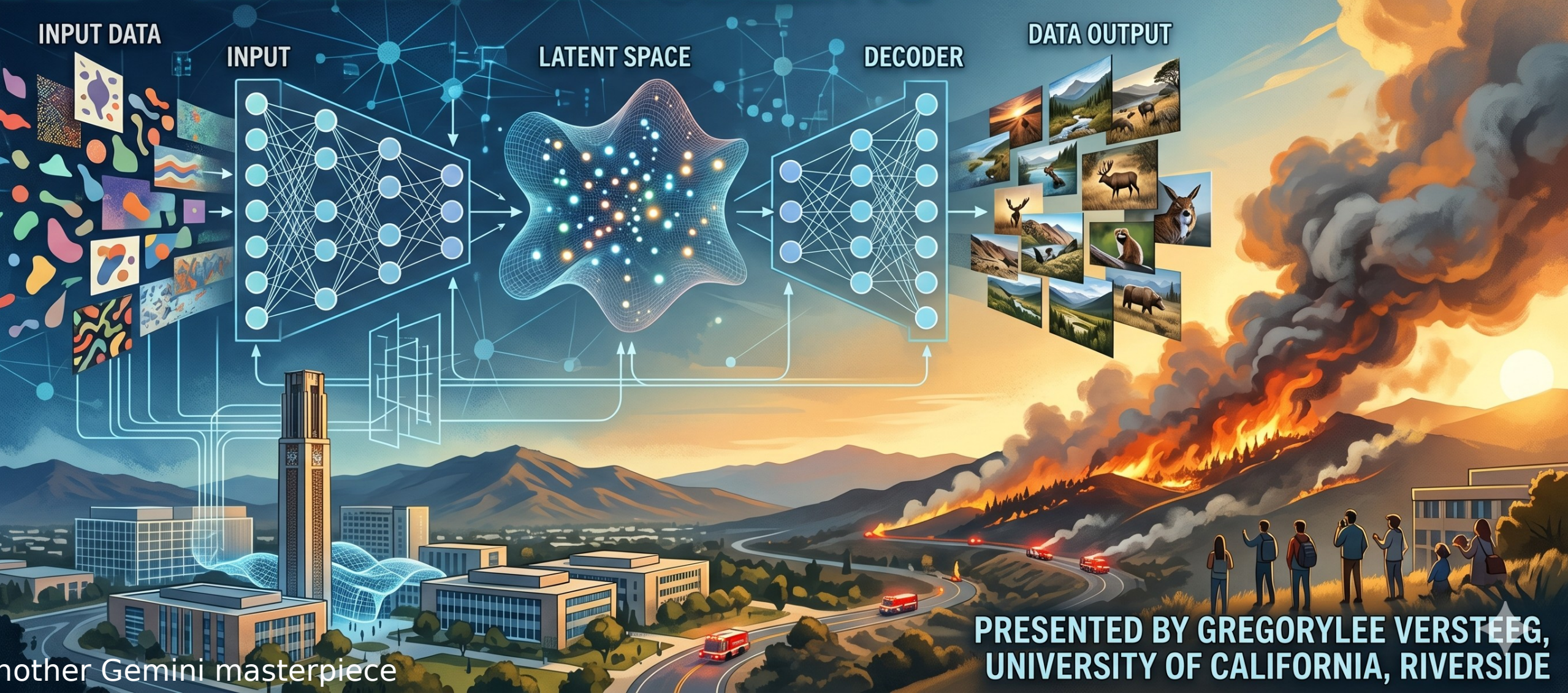
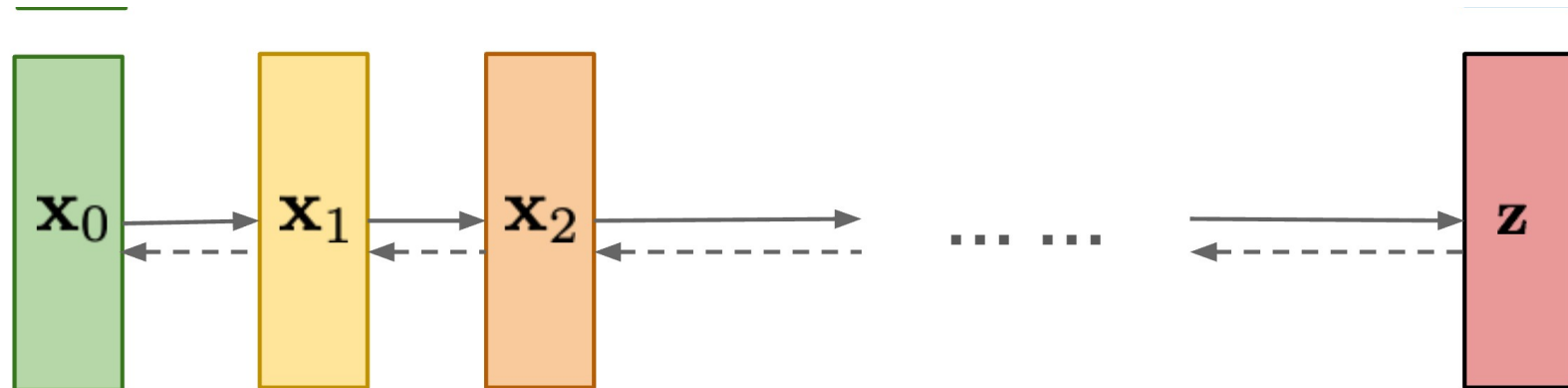


UNVEILING VARIATIONAL AUTOENCODERS (VAEs): DEEP GENERATIVE MODELING

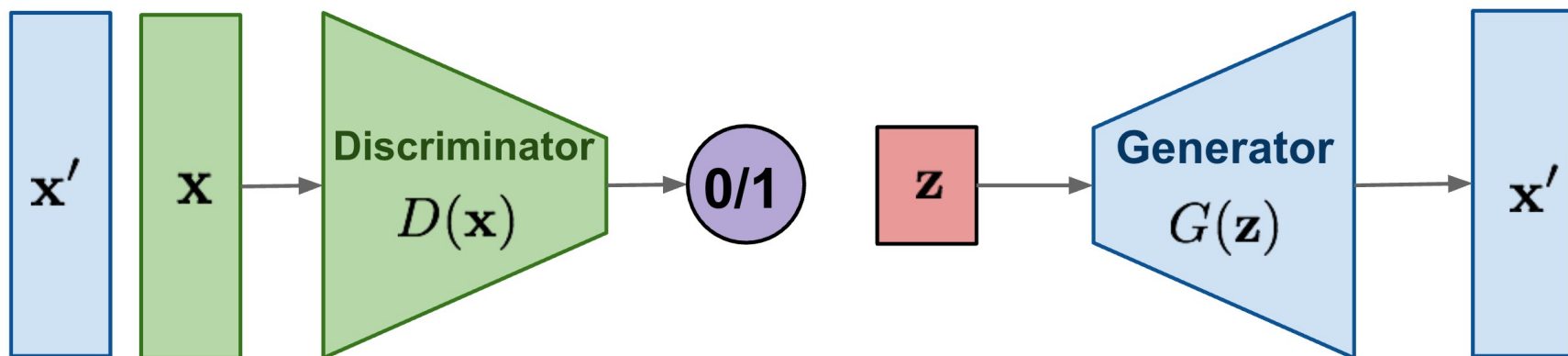


another Gemini masterpiece

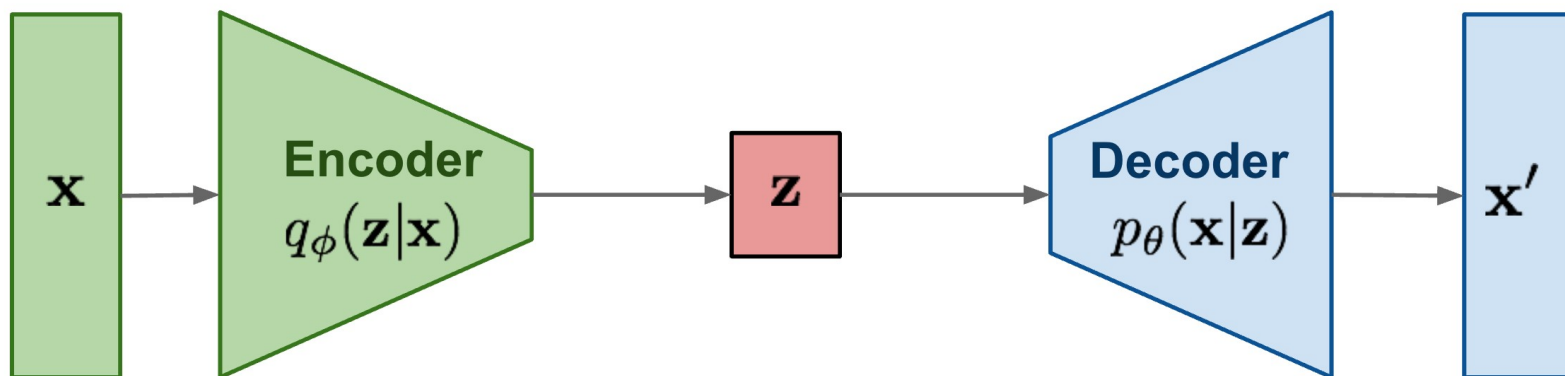
Diffusion models:
Gradually add Gaussian
noise and then reverse

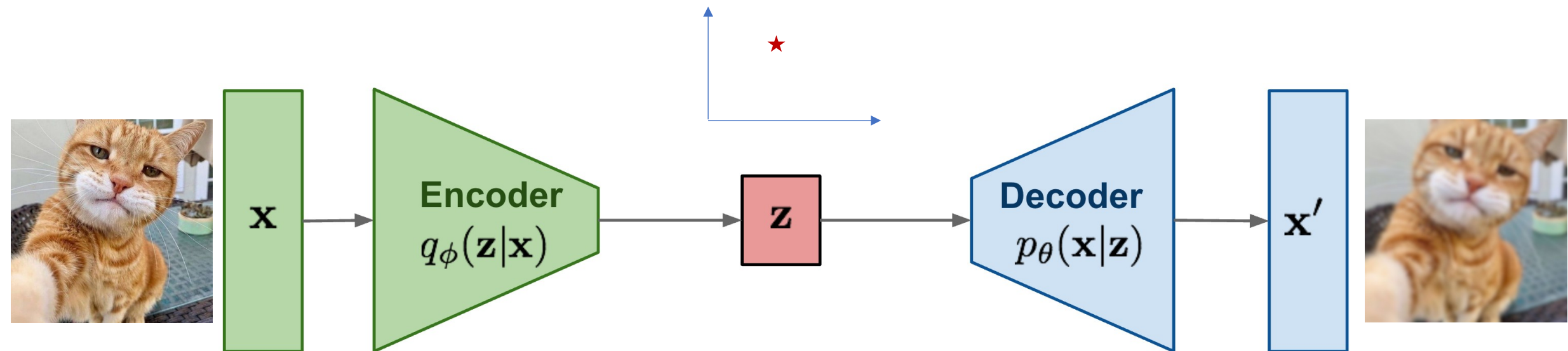


GAN: Adversarial
training



VAE: maximize
variational lower bound





- General autoencoder question: What should the loss function be?
- We're using probability distributions for the encoder-decoder...
 - What does it mean?
 - How to represent them?
 - How to *optimize* them?

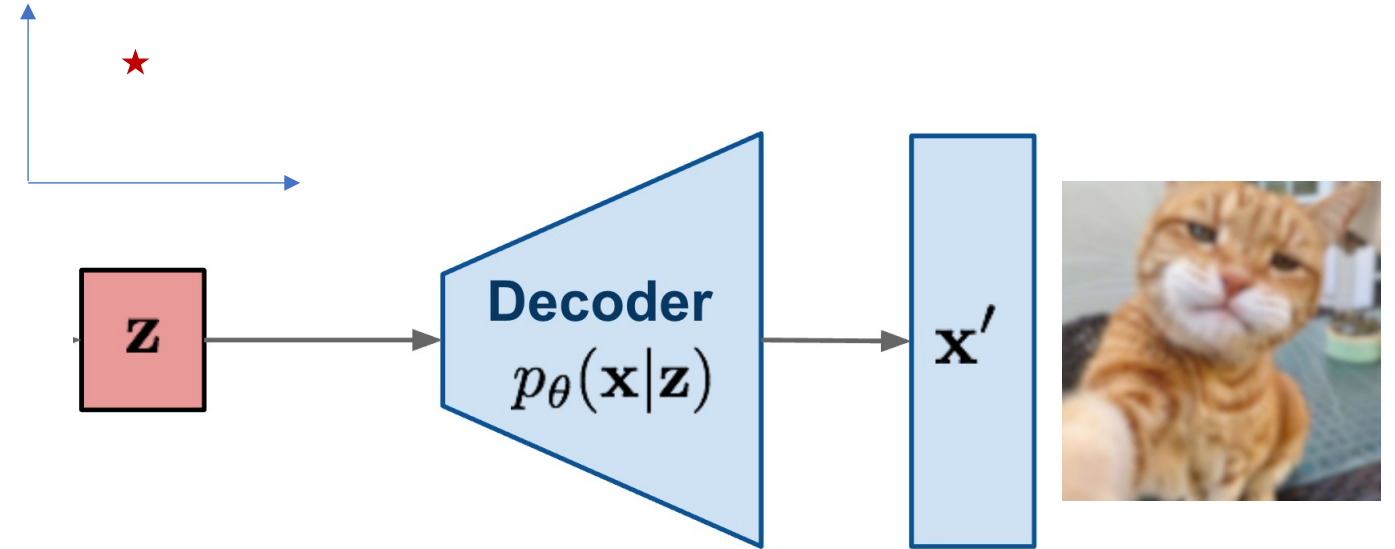
Why Variational Auto-Encoders (VAE)?

Many useful ideas:

- “Variational distribution”
- Evidence Lower BOund (**ELBO**)
- “Reparametrization trick”

VAE and VQ-VAE still used for dimensionality reduction in latent diffusion (and probably for many other things, like audio)

A generative latent factor model (vs GAN)



(Draw on board)

A “generative” latent factor model

$$p_{(\underline{\theta})}(x, z) = p_{\theta}(x|z)p(z)$$

$$\begin{array}{c} Z \quad p(z) \\ \downarrow \\ X \quad p_{\theta}(x|z) \end{array}$$

Example choices:

$$p(z) = \mathcal{N}(z; 0, \mathbf{I})$$

$$p(x|z) = N(x; \mu(z), I)$$

Variational optimization: make the data look likely under the generative model

- Take $x \sim q(x)$ (the data distribution)

$$\max_{\theta} E_{q(x)} \log p_{\theta}(x)$$

- What is $p(x)$?

$$\max_{\theta} 1/N \sum_{\ell=1}^N \log p_{\theta}(x^{(\ell)})$$

It's a hard integral...

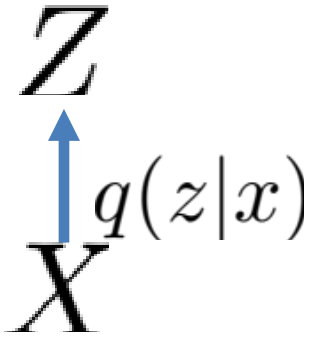
Need many “z” values to estimate,
Each requires a neural net call

$$p(x) = \int dz \, p(x|z)p(z)$$

Bayes rule hint for how to get $p(x)$?

- What if we use a *variational* approximation, $q(z|x)$?

$$D[q(z|x) || p(z|x)] = \mathbb{E}_{q(z|x)} \log \frac{q(z|x)}{p(z|x)} \geq 0$$



One line ELBO derivation

The “gap” in the ELBO

True posterior

$$\log p(x) \geq \log p(x) - D_{KL}[q(z|x) || p(z|x)]$$

Variational approximation

Show actual steps on board

$$= \mathbb{E}_{q(z|x)} [\log p(x|z)] - D_{KL}[q(z|x) || p(z)]$$

Evidence Lower Bound (ELBO)

$$\log p(x) \geq \mathbb{E}_{q(z|x)} [\log p(x|z)] - D[q(z|x) || p(z)]$$

Our
favorite!

Looks like a
probabilistic auto-
encoder
reconstruction!

WTF!

For more about the ELBO: "Variational Inference: A Review for Statisticians": [arXiv:1601.00670](https://arxiv.org/abs/1601.00670)

More complex bounds with extended state space dynamics

$$\log p(x) \geq \log p(x) - D_{KL} [q^{\text{PROP}}(z_{\text{ext}}|x) || p^{\text{TGT}}(z_{\text{ext}}|x)]$$

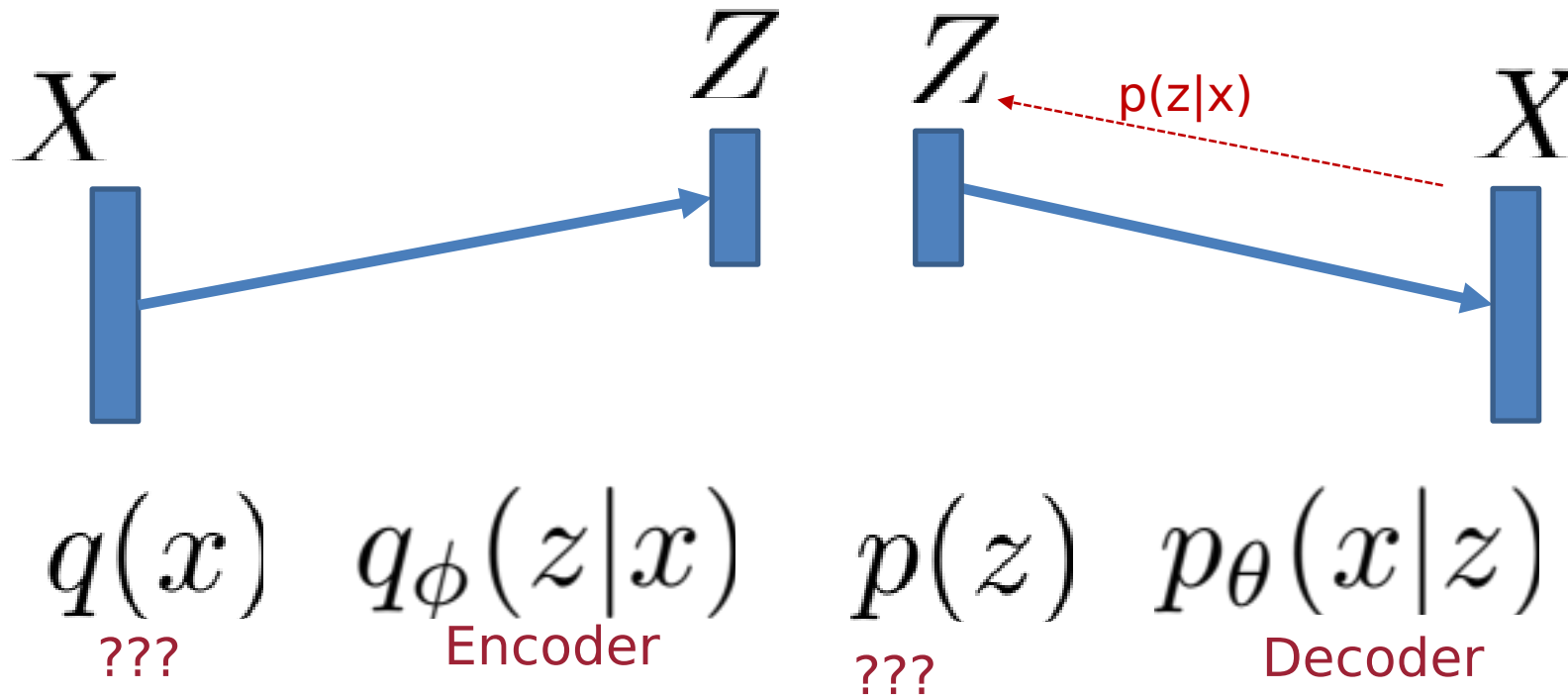
$$\mathbb{E}_{q^{\text{PROP}}(z_{\text{ext}}|x)} \left[\log \frac{p^{\text{TGT}}(x, z_{\text{ext}})}{q^{\text{PROP}}(z_{\text{ext}}|x)} \right]$$

Sample crazy dynamics,
multiple samples per
point

Use tricks in here, to get importance
weighted autoencoder bounds, AIS bounds,
and generalizations

Back to regular VAE

$$\log p(x) \geq \mathbb{E}_{q(z|x)} [\log p(x|z)] - D[q(z|x)||p(z)]$$



Data distribution,
The expression
above works for a
single x

$$\log p(x) \geq \mathbb{E}_{q(z|x)} [\log p(x|z)] - D[q(z|x) || p(z)]$$

- Kingma & Welling call $p(z)$ a “prior” and the encoder regularizes $q(z|x)$ to be as close to the prior as possible.
- We’ll see later that it can also be interpreted in terms of *compression*
- By choosing the encoder, prior, we can get an analytic form for this.

$$q_\phi(z_j|x) = \mathcal{N}(z_j; g_{j,\phi}(x), \text{diag } s_{j,\phi}^2(x))$$

$$p(z) = \mathcal{N}(z; 0, \mathbf{I})$$

- KL divergence between two Gaussians is easy.

$$\log p(x) \geq \mathbb{E}_{q(z|x)} [\log p(x|z)] - D[q(z|x)||p(z)]$$

$$q_{\phi}(z_j|x) = \mathcal{N}(z_j; g_{j,\phi}(x), \text{diag } s_{j,\phi}^2(x))$$

$$p(z) = \mathcal{N}(z; 0, \mathbf{I})$$

KL divergence between two (diagonal) Gaussians is easy.

$$- \frac{1}{2} \sum_{j=1}^m \left(1 + \log s_j(x)^2 - \frac{g_j(x)^2}{s_j(x)^2} \right)$$

$$\log p(x) \geq \underbrace{\mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x|z)]}_{\text{Evidence Lower Bound}} - D[q(z|x) || p(z)]$$

How do we do gradients with respect to phi?

REINFORCE gradient estimator (if time)

$$\nabla_{\theta} \mathbb{E}_{x \sim p_{\theta}(x)} [f(x)]$$

$$\nabla_{\theta} p_{\theta}(x) = p_{\theta}(x) \nabla_{\theta} \log p_{\theta}(x)$$

From this, we can derive the REINFORCE estimator

$$\nabla_{\theta} \mathbb{E}_{x \sim p_{\theta}(x)} [f(x)] = \mathbb{E}_{x \sim p_{\theta}(x)} [f(x) \nabla_{\theta} \log p_{\theta}(x)]$$

$$\log p(x) \geq \mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x|z)] - D[q(z|x) || p(z)]$$

- How do we do gradients with respect to ϕ ? REINFORCE, or
- We can do the “re-parametrization trick”

$$\mathbb{E}_{q_\phi(z|x)} [h(z)] = \mathbb{E}_{q(\epsilon)} [h(z_\phi(x, \epsilon))]$$

- Let's look at a concrete example

Direct example of re-parametrization trick (simplified)

$$\mathbb{E}_{z \sim q_\phi(z|x)}[h(z)] = \mathbb{E}_{\epsilon \sim \mathcal{N}(0,1)}[h(g_\phi(x) + s_\phi(x)\epsilon)]$$

Direct example of re-parametrization trick

$$\mathbb{E}_{q_{\phi}(z|x)}[h(z)] = \int dz \frac{e^{-1/2 \left(\frac{z - g_{\phi}(x)}{s_{\phi}(x)} \right)^2}}{\sqrt{2\pi s_{\phi}(x)^2}} h(z)$$

Change of
variables
 $\epsilon = (z - g_{\phi}(x))/s_{\phi}(x)$
 $z = g_{\phi}(x) + s_{\phi}(x)\epsilon$
 $d\epsilon = dz/s_{\phi}(x)$

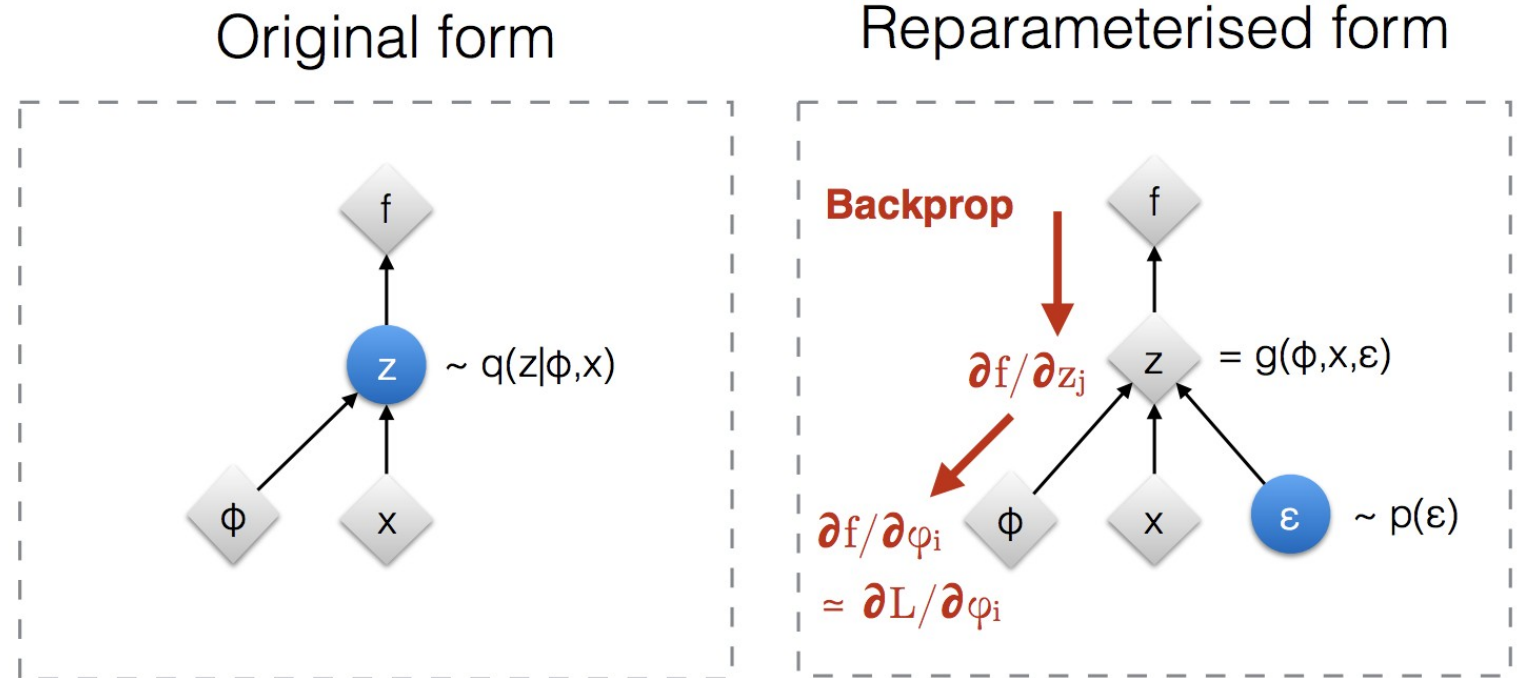
$$= \int d\epsilon \frac{e^{-\epsilon^2/2}}{\sqrt{2\pi}} h(g_{\phi}(x) + s_{\phi}(x)\epsilon)$$

$$= \mathbb{E}_{\epsilon \sim \mathcal{N}(0,1)}[h(g_{\phi}(x) + s_{\phi}(x)\epsilon)]$$

$$\approx 1/L \sum_{\ell=1}^L h(g_{\phi}(x) + s_{\phi}(x)\epsilon_{\ell})$$

Visual depiction of re-parametrization

<https://lilianweng.github.io/lil-log/2018/08/12/from-autoencoder-to-beta-vae.html>



◆ : Deterministic node
● : Random node

[Kingma, 2013]
[Bengio, 2013]
[Kingma and Welling 2014]
[Rezende et al 2014]

When can you use re-parametrization trick?

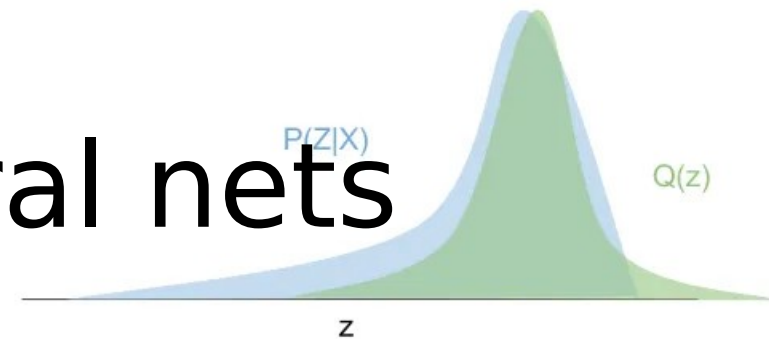
$$\mathbb{E}_{q_{\phi}(z|x)}[h(z)] = \mathbb{E}_{q(\epsilon)}[h(z_{\phi}(x, \epsilon))]$$

- You can do this for many (continuous) distributions – Kingma & Welling list examples
- You can even do this with discrete random variables using the Gumbel softmax trick:

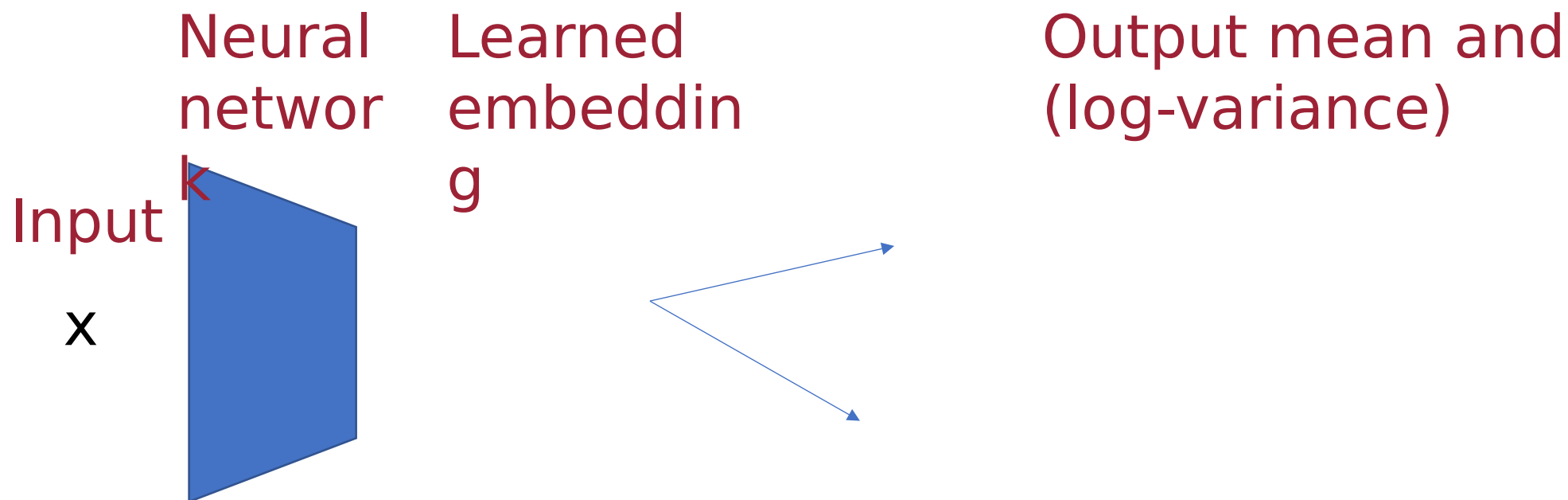
<https://arxiv.org/pdf/1611.01144.pdf>

Implementation details

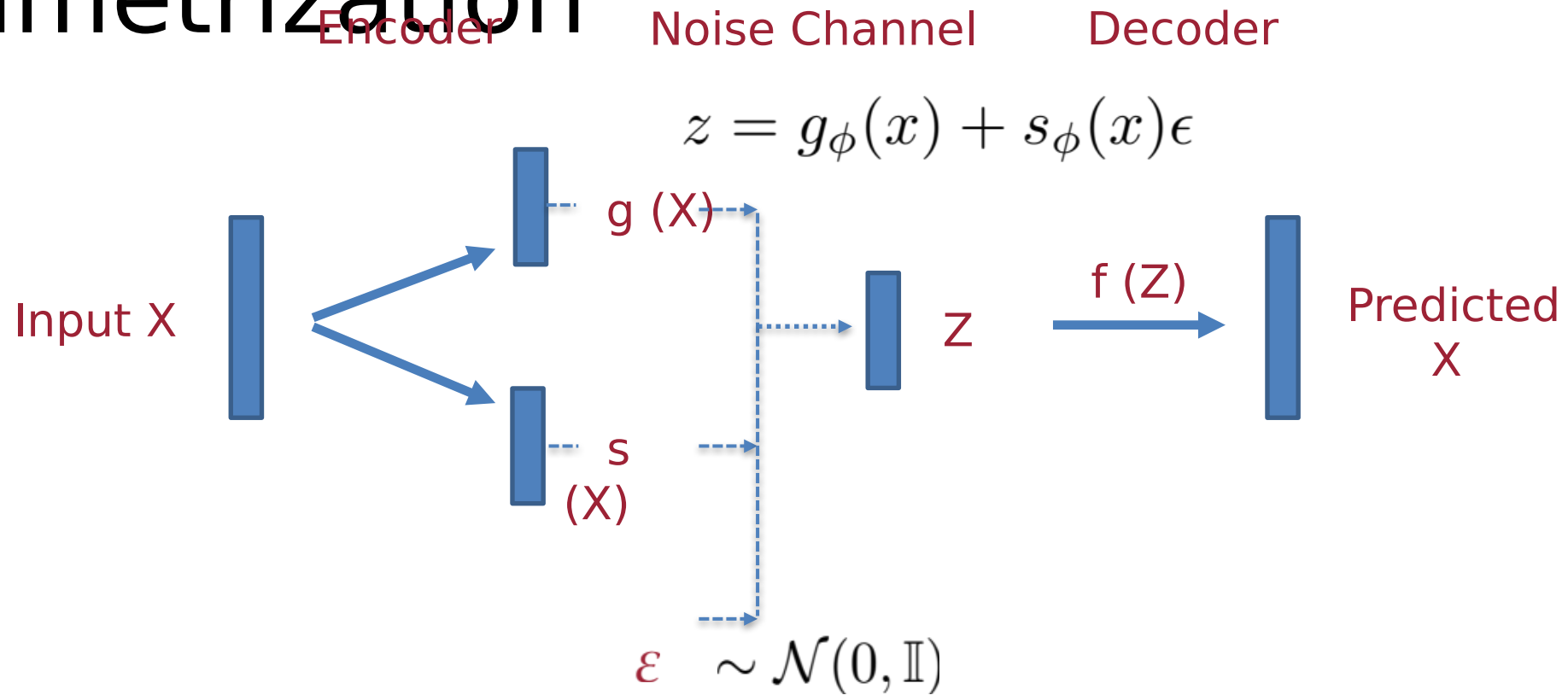
Variational inference neural nets



-



Visualizing architecture with re-parametrization



Minimize (Negative) Evidence Lower Bound (ELBO)

$$-\log p(x) \leq \underbrace{-\mathbb{E}_{q(z|x)} [\log p(x|z)]}_{\text{reconstruction}} + \underbrace{D[q(z|x) || p(z)]}_{\text{regularization}}$$

$$\approx -\log p(X = x | Z = g_\phi(x) + s_\phi(x)\epsilon) - 1/2 \sum_{j=1}^m (1 + \log s_j(x)^2 - g_j(x)^2 - s_j(x)^2)$$

What *exactly* is term on left?

Reconstruction with continuous random variables

In that case, a common choice is to use:

$$p(x|z) = \mathcal{N}(f_{\theta}(z), I)$$

Because then $-\log p(x|z)$ becomes:

$$-\log p(x|z) = 1/2 \|x - f(z)\|_2^2 + d/2 \log(2\pi)$$

The mean square error!

- $f(z)$ is actually the “mean decoding”
- Nice that we can compare to deterministic autoencoders – same loss plus a regularizer. Also easy to specify in pytorch

What can we do with it? (computational graph surgery)

Reconstruction

Encoder

Noise Channel

Decoder

$\log p(x) \geq$ ELBO Loss

Estimate likelihood

Use z's for something

Generate:

Use your own z's

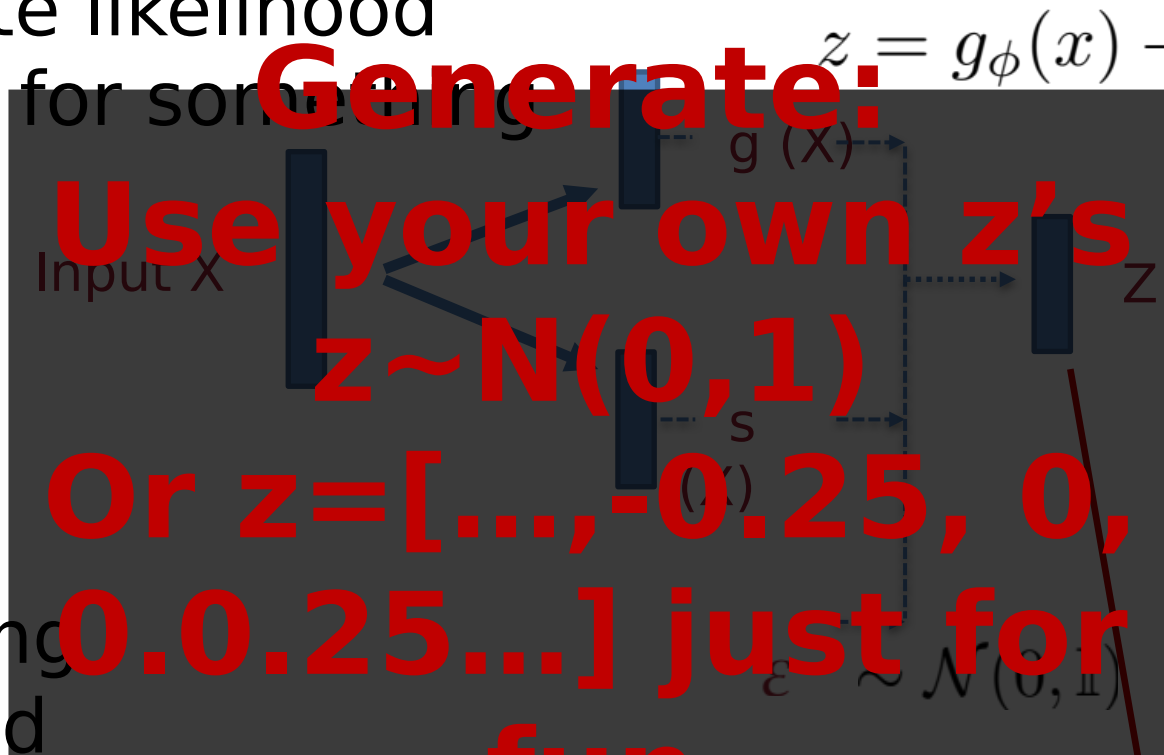
$z \sim \mathcal{N}(0, 1)$

Or $z = [\dots, -0.25, 0,$

$0.0.25 \dots]$ just for

fun

Sampling
Manifold



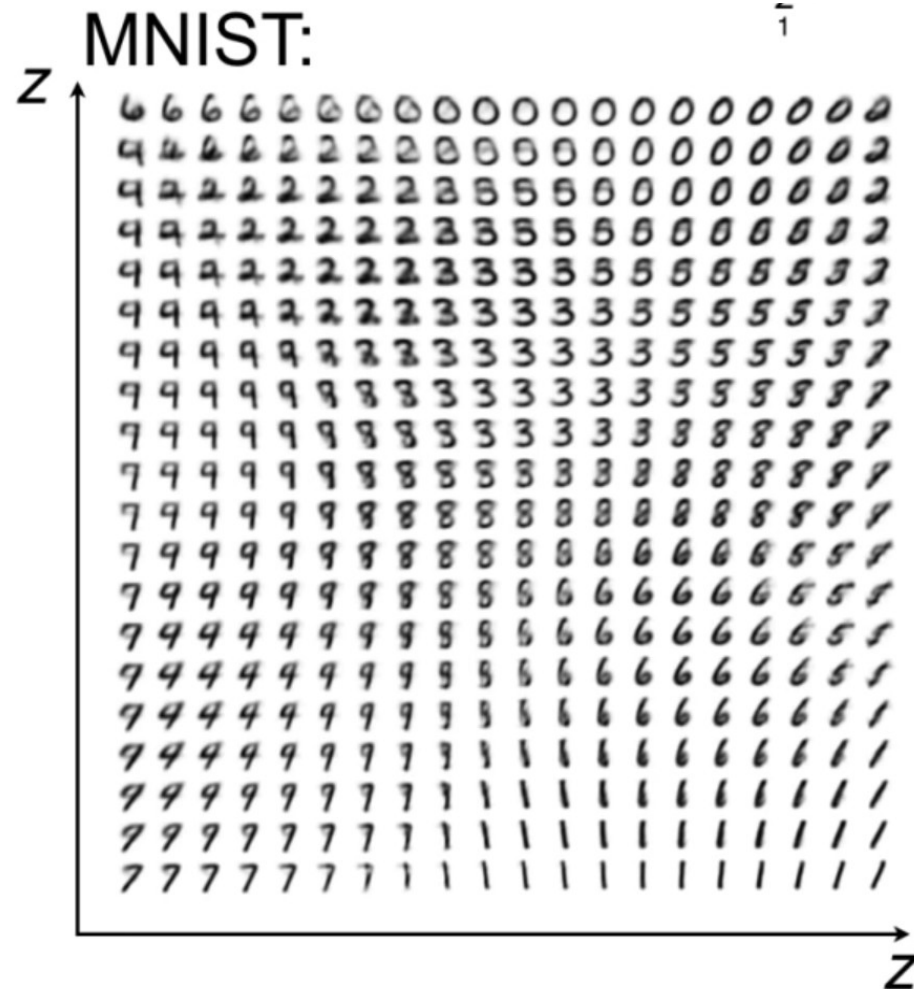
$$z = g_{\phi}(x) + s_{\phi}(x)\epsilon$$

$f(z)$

Predicted
X

Dimensionality

That's how we get this plot – show the mean reconstruction for each value of z

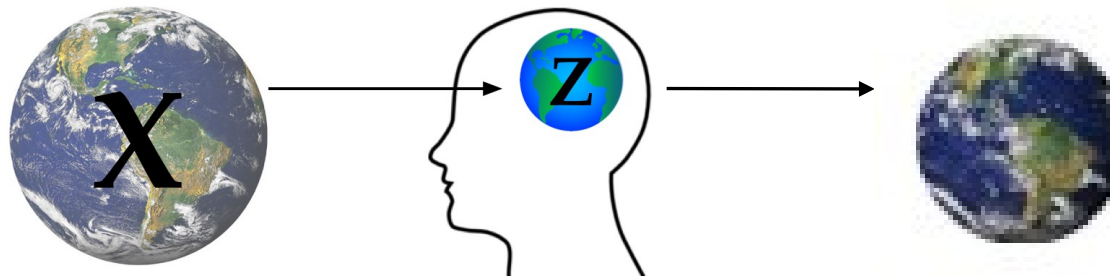


VAE Recap

-

$$p(x) = \int dz \, p(x|z)p(z)$$

$$-\log p(x) \leq \underbrace{-\mathbb{E}_{q(z|x)} [\log p(x|z)]}_{\text{reconstruction}} + \underbrace{D[q(z|x)||p(z)]}_{\text{regularization}}$$



Z is a compressed *representation* of X, that should be useful for reconstruction, β controls compression

Rate-distortion / lossy compression formulation of VAE

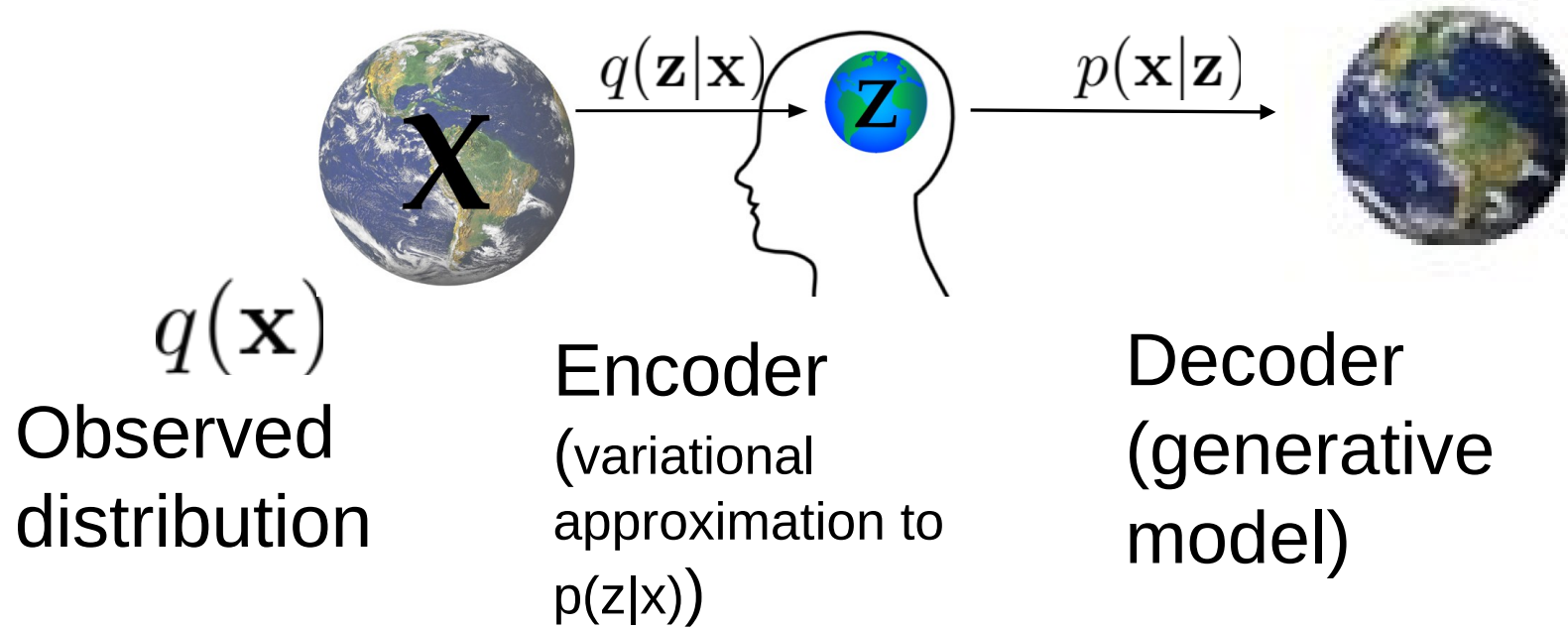
Rate-distortion connection is discussed in several papers:

Alemi, A., Poole, B., Fischer, I., Dillon, J., Saourous, R. A., and Murphy, K. Fixing a broken ELBO. ICML 2018

Rezende, D. J. and Viola, F. Taming VAEs. *arXiv preprint arXiv:1810.00597*, 2018.

Brekelmans, Moyer, Galstyan, and Ver Steeg. "Exact Rate-Distortion in Autoencoders via Echo Noise." *NeurIPS 2019*

Variational Auto-Encoder (VAE)



- Encoder and decoder parametrized by neural nets
- Goal is to maximize likelihood of observations under generative model

The Rate-Distortion Problem

- Min rate for given distortion level D

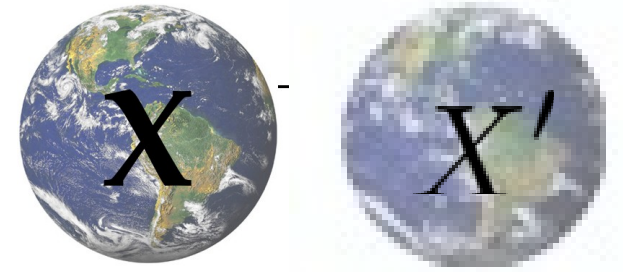
$$R(D) = \min I_q(X; Z)$$

$$\text{subject to: } \mathbb{E}[d(x, x')] \leq D$$

- Unconstrained optimization

$$\min I(X; Z) + \beta \mathbb{E}[d(x, x')]$$

- (beta just controls the ratio of these two terms)



$d()$ is a distortion
measure on x, x'

Minimize (Negative) Evidence Lower Bound (ELBO)

$$-\log p(x) \leq \underbrace{-\mathbb{E}_{q(z|x)} [\log p(x|z)]}_{\text{Reconstruction ("distortion")}} + \underbrace{\beta D[q(z|x) || p(z)]}_{\text{Rate (upper bound)}}$$

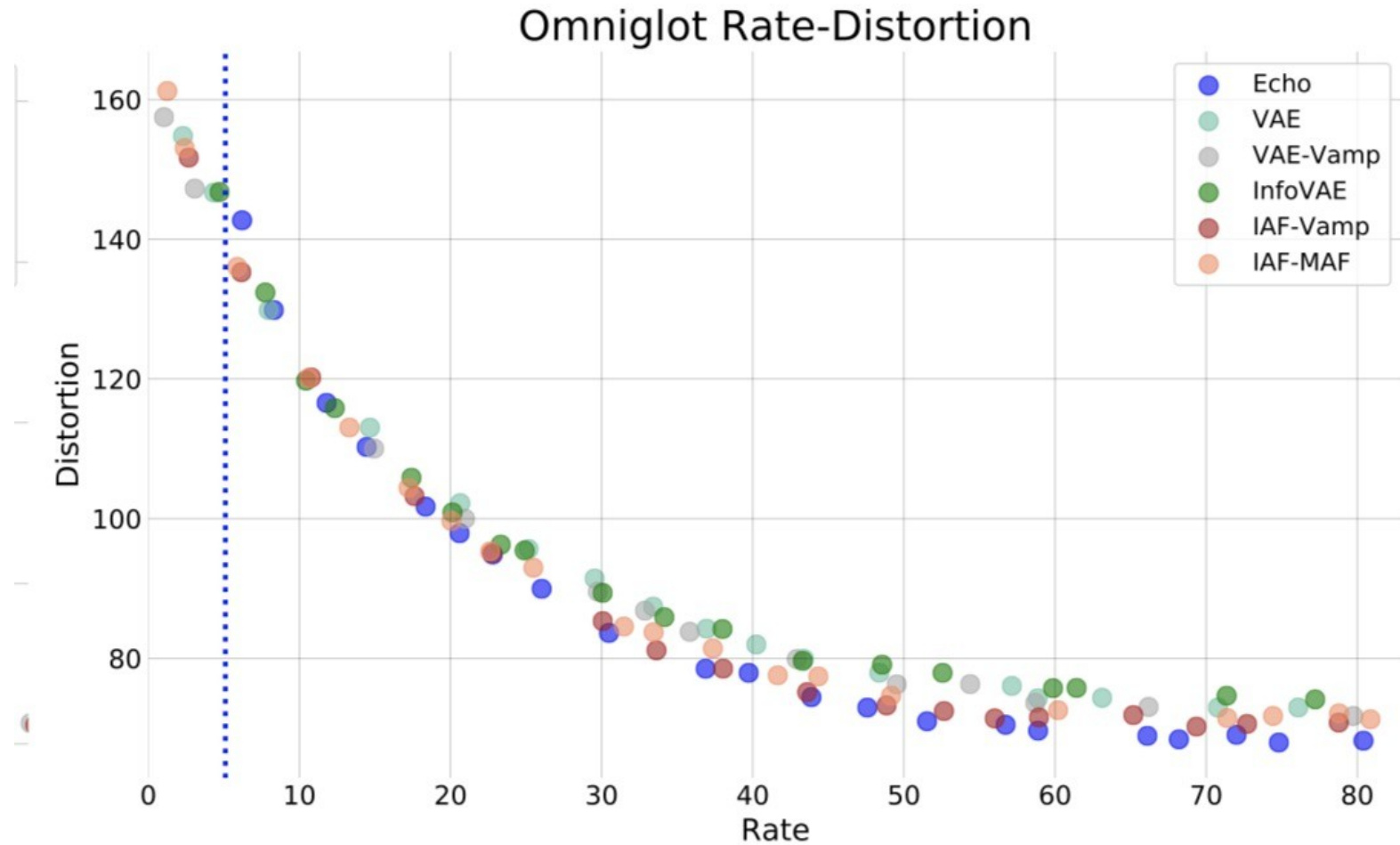
$$\mathbb{E}_{q(x)}[D[q(z|x) || p(z)]] = E_{q(x)}[D[q(z|x) || q(z)] + D[q(z) || p(z)]] \geq E_{q(x)}[D[q(z|x) || q(z)]] = I(z; x)$$

Choose $p(z) = q(z) = \int q(z|x)q(x)dx$ for tightest bound

By adding “beta” ([beta VAE](#)) you can control the trade-off.

Rate-distortion curve

- I took this example from Brekelmans et al NeurIPS 2019
- Different points on the curve are achieved by varying beta.
- Beta=1 is special – it gives us the ELBO, a lower bound on log likelihood.
- Achievable points depend on a lot of things – architecture, optimizer, data
- More details like theoretical bounds discussed in Alemi:
<https://arxiv.org/pdf/1711.00464.pdf>



Visualizing rate-distortion trade-offs on OmniGlot

Original

0 H K W C P

$\beta = 2.0$
R=18.3 D=101.7

H K W C P

$\beta = 1.0$
R=30.5 D=83.6

0 H K W C P

$\beta = 0.1$
R=72.0 D=69.0

0 H K W C P

Echo

$\beta = 1.75$
R=20.1 D=100.9

H K W C P

$\beta = 1.0$
R=30.1 D=89.4

0 H K W C P

$\beta = 0.15$
R=71.4 D=74.6

0 H K W C P

VAE

Vector Quantized-VAE

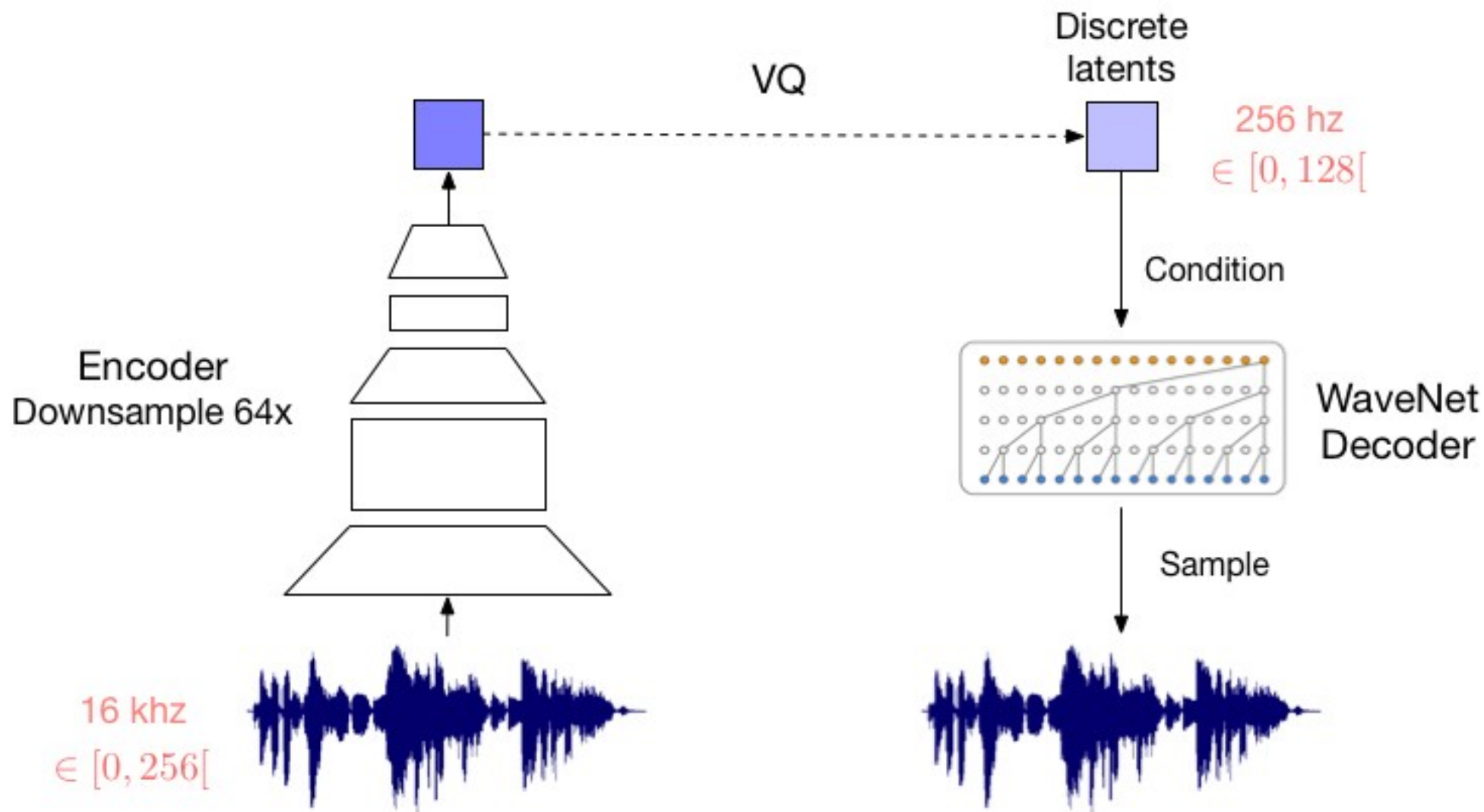
Fixing the rate via *quantization*

<https://mlberkeley.substack.com/p/vq-vae>

<https://avdnoord.github.io/homepage/vqvae/>

<https://arxiv.org/abs/1711.00937>

Discrete "codebooks" rather than Gaussian "z" is more natural if you want to store a compressed version!



How do we do this with a VAE?

Encoder



image to
discrete codes



56	73	67	23	81	19	...
----	----	----	----	----	----	-----

Decoder

56	73	67	23	81	19	...
----	----	----	----	----	----	-----

discrete codes
to image

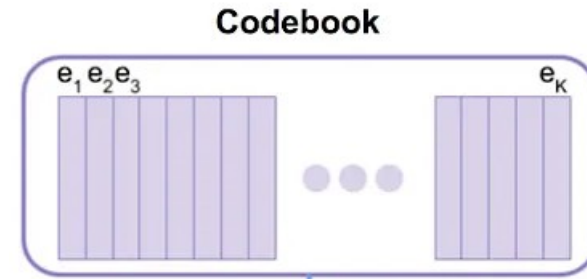
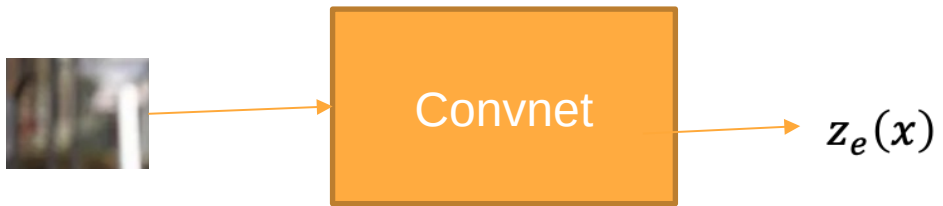


We had “reconstruction” plus “rate”.

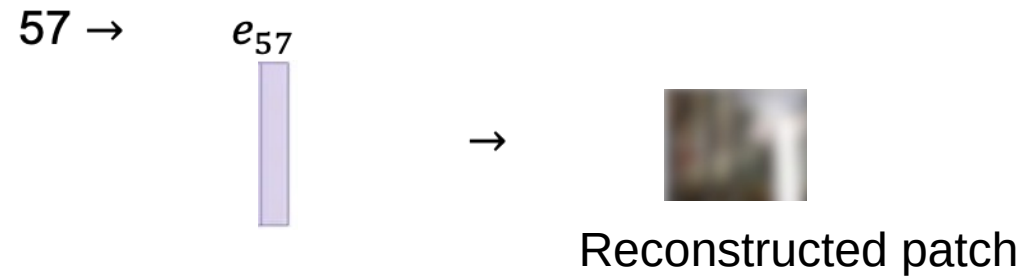
The rate is now capped by the discrete representation!

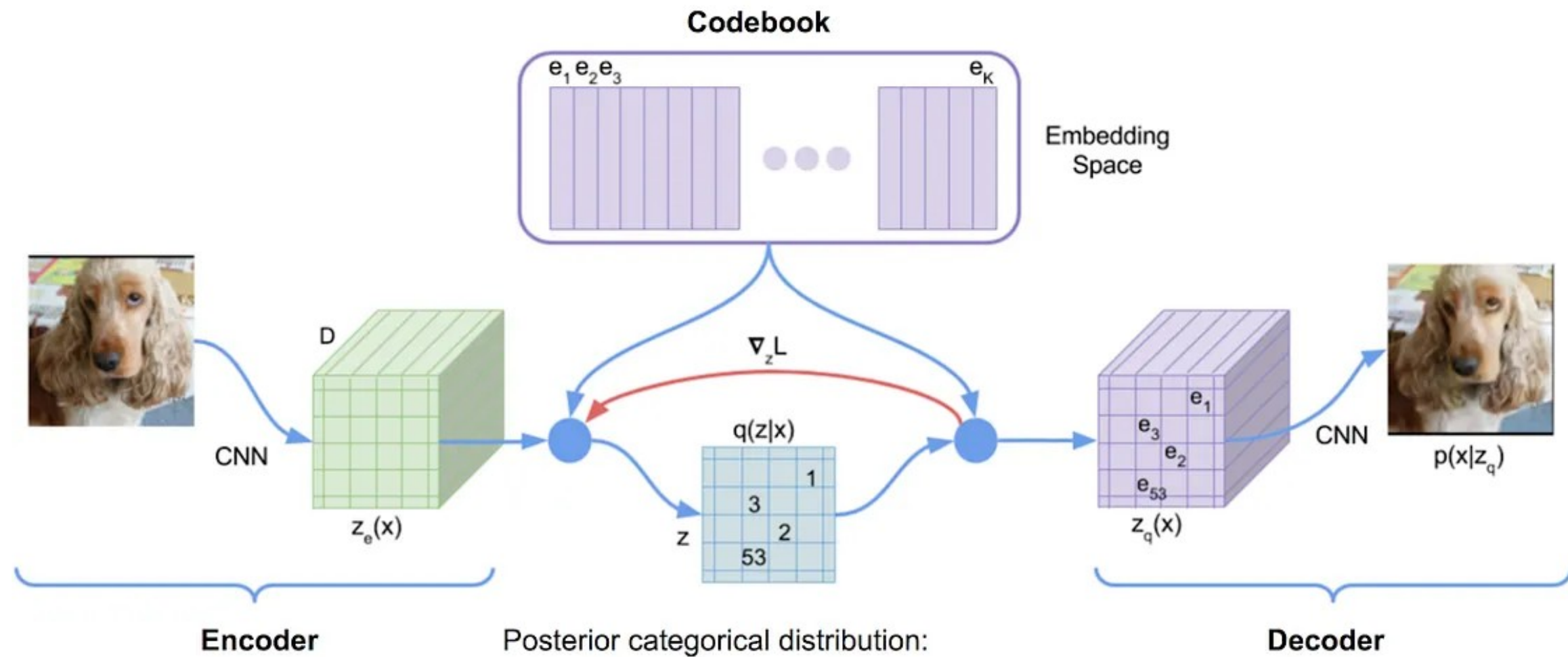
So we just need to encourage good reconstruction!

Codebook



- Find closest vector for this patch
- Encode as the index "57", e.g.





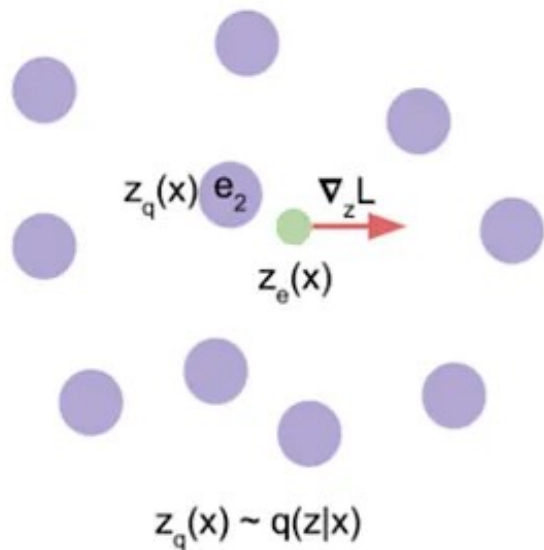
Loss function for training

$$L = \log p(x|z_q(x)) + \|\text{sg}[z_e(x)] - e\|_2^2 + \beta \|z_e(x) - \text{sg}[e]\|_2^2,$$

Reconstruction

Optimize dictionary,
so codes match
output of
continuous net

Optimize encoder,
to give outputs
close to codes
("commitment loss")

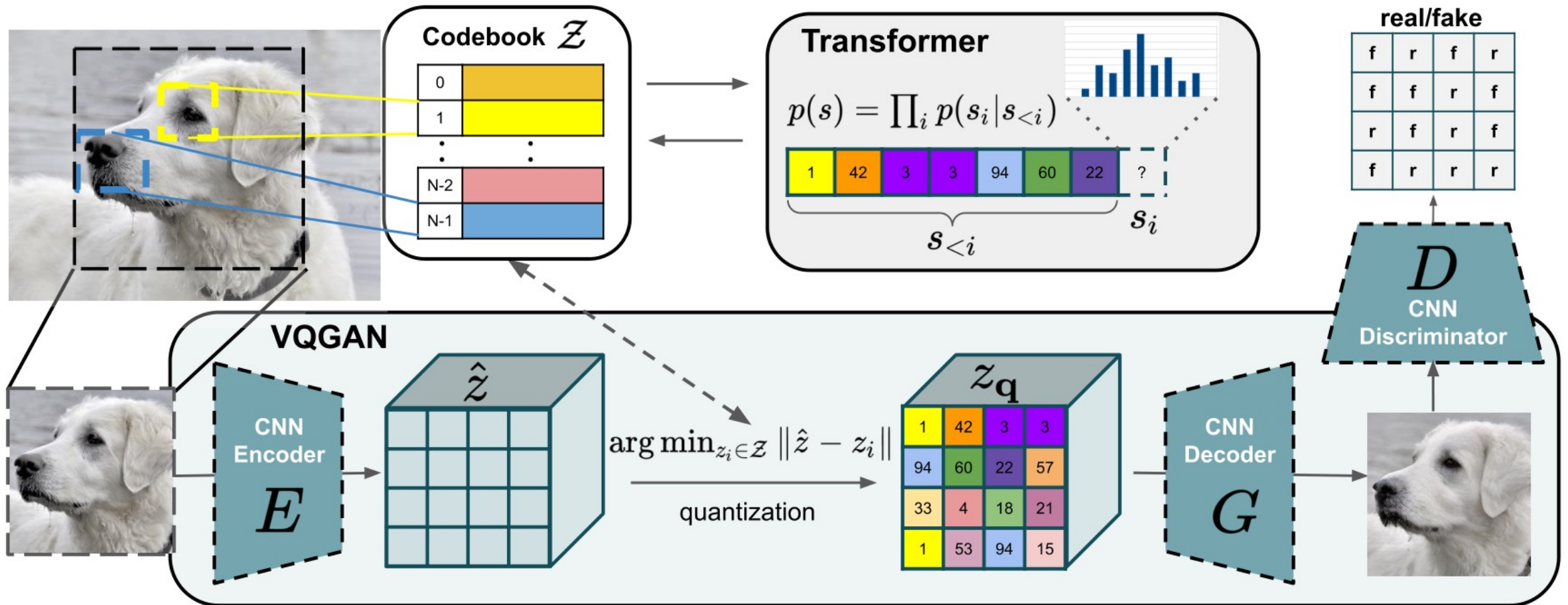


Make sure that continuous embedding is close to discrete embedding

Vector Quantized: VQ-GAN

<https://compvis.github.io/taming-transformers>

Identical to VQ-VAE, but we add the GAN loss as a regularizer!



Monday: Holiday (Memorial day)